

1- IMPORT LIBRARIES

```
[3]
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2- LOAD DATASET

```
[4]
# Load the dataset (.csv) file ---
df = pd.read_csv('netflix_movie_tvshow.csv')
df.head()
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mablane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Samir Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train l...

Next steps: [Generate code with df](#) [New interactive sheet](#)

- categorical - type, title, director, country, cast, rating, listed_in
- numerical - released_year
- date - date_added
- mix- show_id, description, duration,

3- EXPLORATION OF THE DATASET

```
[5]: df.shape # >> we have 8887 rows and 12 columns <<
df.columns
df.values
df.dtypes
```

```
***
0
show_id    object
type       object
title      object
director   object
cast       object
country    object
date_added object
release_year  int64
rating     object
duration   object
listed_in  object
description object

dtype: object
```

```
[5]: # INFO ABOUT THE DF LIKE --- TOTAL NUMBER OF ROWS, TOTAL NUMBER OF COLUMNS, DATATYPES OF EACH COLUMN AND MEMORY REQUIREMENT ---
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8887 entries, 0 to 8886
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         8887 non-null  object
1   type            8887 non-null  object
-   ...            ...          ...
-   ...            ...          ...
```

```
[6]
✓ Os # INFO ABOUT THE DF LIKE --- TOTAL NUMBER OF ROWS, TOTAL NUMBER OF COLUMNS, DATATYPES OF EACH COLUMN AND MEMORY REQUIREMENT ---
df.info()
```

```
...
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   show_id     8807 non-null   object
1   type        8807 non-null   object
2   title       8807 non-null   object
3   director    6173 non-null   object
4   cast        7982 non-null   object
5   country     7976 non-null   object
6   date_added  8797 non-null   object
7   release_year 8807 non-null   int64
8   rating      8803 non-null   object
9   duration    8804 non-null   object
10  listed_in   8807 non-null   object
11  description  8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

```
[7]
✓ Os # TOP 10 ROWS OF THE DF ---
df.head(10)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
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5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gifford, Hamish Linklater, H...	NaN	September 24, 2021	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries	The arrival of a charismatic young priest brin...
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	NaN	September 24, 2021	2021	PG	91 min	Children & Family Movies	Equestria's divided. But a bright-eyed hero be...

```
[9]: df.describe()
```

↑ ↓ ✎ 🗑 ⋮

*** release_year 📊

count	8807.000000
mean	2014.180198
std	8.818312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

4- DATA CLEANING

```
[10]: # CHECK MISSING VALUES IN THE DF ---
df.isnull().sum() # >>> Total no of missing values in each column <<<
```

	0
show_id	0
type	0
title	0
director	2834
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
studio	0

```
[11] # CHECK FOR DUPLICATE VALUES ---
✓ Os df.duplicated().sum() # >>> we found no duplicates in this df <<<

np.int64(0)
```

```
[12]
✓ Os df['director'].fillna('Unknown', inplace=True)
df['cast'].fillna('Not Available', inplace=True)
df['country'].fillna('Not Specified', inplace=True)

*** /tmp/ipykernel_21316/727864186.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

    df['director'].fillna('Unknown', inplace=True)
/tmp/ipykernel_21316/727864186.py:2: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

    df['cast'].fillna('Not Available', inplace=True)
/tmp/ipykernel_21316/727864186.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

    df['country'].fillna('Not Specified', inplace=True)
```

```
[13]
✓ Os # DROPPING THE ROWS WHICH WE CANT REPLACE EITHER WITH MEAN AND MEDIAN THEY ARE TOO LESS IN NUMBERS TO AFFECT THE ANALYSIS ---
df.dropna(subset= ['rating', 'duration'], inplace=True)
df.dropna(subset= ['date_added'], inplace=True)
```

```
[14]
✓ Os df.to_csv('cleaned_netflix_movie_tvshow')
```

5- Data Analysis

```
[15]
✓ Os df['type'].value_counts() # >>> there are 6126 'Movies' and 2664 'TV shows'
```

5- Data Analysis

```
[15] df['type'].value_counts() # >>> there are 6126 'Movies' and 2664 'TV shows'
```

```
count
type
Movie    6126
TV Show   2664
dtype: int64
```

```
[16] df.loc[df['type'] == 'Movie'].head() # >>> select only Movies using label-based filtering <<<
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	Not Available	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	Not Specified	September 24, 2021	2021	PG	91 min	Children & Family Movies	Equestria's divided. But a bright-eyed hero be...
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	September 24, 2021	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies	On a photo shoot in Ghana, an American model s...
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	September 24, 2021	2021	PG-13	104 min	Comedies, Dramas	A woman adjusting to life after a loss contend...
12	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...	Germany, Czech Republic	September 23, 2021	2021	TV-MA	127 min	Dramas, International Movies	After most of her family is murdered in a terr...

```
[17] # Count of Movies vs TV Shows per country (top 5 countries)
exclude = ['Unknown', 'Not Available', 'Not Specified']

top_countries = df['country'].value_counts().head(6).index
filtered = df[(df['country'].isin(top_countries)) & (~df['country'].isin(exclude))]

filtered.groupby(['country', 'type'])['title'].count()
```

title

```
[17] ✓ On
# Count of Movies vs TV Shows per country (top 5 countries)
exclude = ['Unknown', 'Not Available', 'Not Specified']

top_countries = df['country'].value_counts().head(6).index
filtered = df[(df['country'].isin(top_countries)) & (~df['country'].isin(exclude))]

filtered.groupby(['country', 'type'])['title'].count()
```

...

		title
country	type	
India	Movie	893
	TV Show	78
Japan	Movie	78
	TV Show	187
South Korea	Movie	41
	TV Show	158
United Kingdom	Movie	208
	TV Show	212
United States	Movie	2055
	TV Show	754

dtype: int64

6 - Visualization

Observation 1 — Countries by Content Count

- United States has the **highest** count with **2809** contents
- Egypt has the **lowest** count with **106** contents only

```
[18] ✓ On
exclude = ['Unknown', 'Not Available', 'Not Specified']
cleaned_country = df[~df['country'].isin(exclude)]['country'].value_counts().head(10) # >>> here i excluded the rows where Country = Not Specified <<<
```

6 - Visualization

Observation 1 — Countries by Content Count

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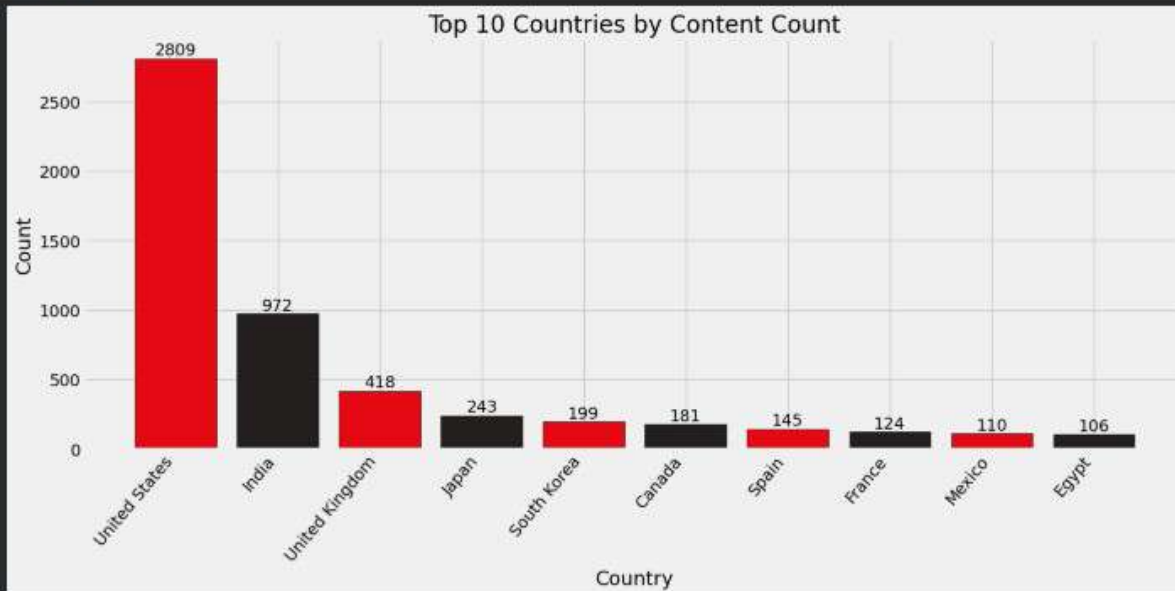
```
exclude = ['Unknown', 'Not Available', 'Not Specified']
cleaned_country = df[df['country'].isin(exclude)]['country'].value_counts().head(10) # >>> here I excluded the rows where Country = Not Specified <<<

plt.figure(figsize=(15, 8))

bar = plt.bar(cleaned_country.index, cleaned_country.values, ec='black', color=['#E58914', '#221F1F'])

plt.xlabel('Country')
plt.ylabel('Count')
plt.title('Top 10 Countries by Content Count')
plt.xticks(rotation=50, ha='right')
plt.bar_label(bar)

plt.show()
```



Country

Observation 2 — Movies vs TV Shows

- Netflix has significantly more **Movies (6,126)** compared to **TV Shows (2,664)**
- Movies make up approximately **70%** of total Netflix content

```
[19] ✓Ds
counts = df['type'].value_counts()

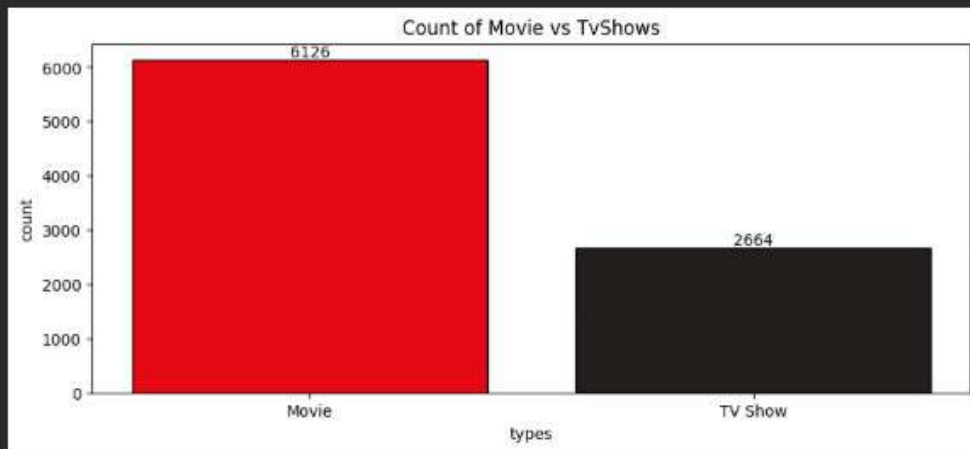
plt.figure(figsize=[ 10,4])

bar = plt.bar(counts.index , counts.values, ec= 'black', color= ['#E58914', '#221F1F'])

plt.title('Count of Movie vs TvShows')
plt.xlabel('types')
plt.ylabel('count')

plt.bar_label(bar)

plt.show()
```



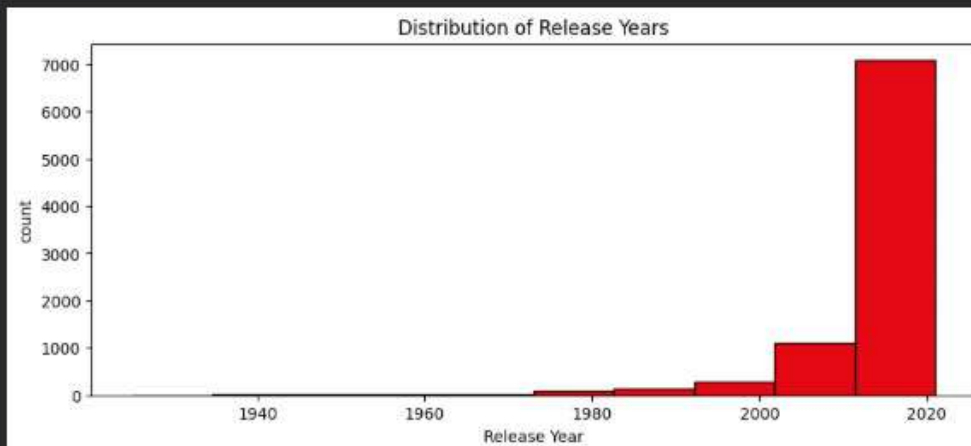
Observation 3 — Distribution of Release Years

- Most of the content on Netflix is from recent years
- Very little content is available from old years like 1940 or 1960

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```
[20]: plt.figure(figsize= [10,4])  
  
plt.hist(df['release_year'], bins= 10, edgecolor= 'black', color= '#E50914')  
plt.xlabel('Release Year')  
plt.ylabel('count')  
plt.title('Distribution of Release Years')  
  
plt.show()
```



Observation 4 — Top 3 Ratings

- TV-MA is the most common rating with over 50% of all content
- This means Netflix mostly has content made for adults
- TV-PG has the least share among the top 3 ratings

```
[21]: top3_rating = df['rating'].value_counts().sort_values(ascending= False).head(3).values  
Rating_label = ['TV-MA', 'TV-14', 'TV-PG']  
  
plt.pie(top3_rating, labels= Rating_label, autopct= '% 0.2f%%', colors= ['#E50914', '#B20710', '#221F1F'], wedgeprops={'edgecolor': 'black'})  
plt.title('Top 3 Content Ratings')  
  
plt.show()
```



Observation 4 — Top 3 Ratings

- TV-MA is the most common rating with over 50% of all content
- This means Netflix mostly has content made for adults
- TV-PG has the least share among the top 3 ratings

```
[21] ▶ top3_rating = df['rating'].value_counts().sort_values(ascending=False).head(3).values
    Rating_label = ['TV-MA', 'TV-14', 'TV-PG']

plt.pie(top3_rating, labels= Rating_label, autopct= '% 0.2f%%', colors= ['#E50914', '#B20710', '#221F1F'], wedgeprops={'edgecolor': 'black'})
plt.title('Top 3 Content Ratings')

plt.show()
```

